

One World or Two: Where Ax1Gen Reads Φ , and Gödel’s Th3 in S4

Drew Arrowood

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Abstract

Benzmüller and Scott leave open whether Theorem Th3 of their actualist Figure 7—possible existence of a God-like being implies necessary existence, with $P(G)$ derived from the generalized conjunction axiom Ax1Gen—is provable in S4. For a Lean 4 transcription of the Archive of Formal Proofs text, with full comprehension, the literal axiom does more than force symmetry. With Ax2a, Ax2b, and Ax4 it forces reflexivity and the B schema $q \supset \Box \Diamond q$. Adding Ax3 forces the accessibility relation to be the identity, and modal collapse follows, with no symmetry hypothesis. Their S5 theory already proves a collapse lemma from symmetry. What is new is the collapse without that hypothesis. The only S4 models of the literal axioms are discrete frames, and Th3 holds there trivially. Two reconstructions, not dictated by the footnote, make both halves of Ax1Gen read Φ at the same world: positivity inside the box, or a world-invariant Φ . Both still yield $P(G)$, neither forces symmetry, and both fail on a two-world S4 chain. Among the three readings examined, the answer turns on whether both halves read Φ at the same world. Domains and Figure 8 were not varied.

1 The open problem

Around 1970 Kurt Gödel wrote down, in a page and a half of symbols, a modal version of the ontological argument [11]. The first axiom says that the conjunction of two positive properties is positive. A footnote extends the axiom from two summands to any number of them: the conjunction of a collection of positive properties is itself positive. Dana Scott, working from a copy Gödel allowed him to make, rearranged the package [14, 15, 16]. In Scott’s notes the generalized conjunction is replaced by a direct postulate that being God-like is positive. Scott also inserts a conjunct into the definition of essence, requiring that the essential property actually hold of the individual, and he takes necessary existence to be positive. That last change is the one that keeps the notes off the inconsistent reading of the unrepaired manuscript.

Jordan Howard Sobel later showed that Scott’s premises collapse modality: whatever is true is necessarily true [15, 16]. The collapse has since been machine-checked, and so have repairs in the spirit of Anderson and of Fitting [1, 10, 5]. Separately, the passage from “possibly God-like” to “necessarily God-like” on the Scott side needs symmetry of accessibility, the modal axiom B, and does not need the rest of S5 [12]. Those facts are recorded again in Section 6 so that they are not mistaken for the theorems below. For Figure 7, collapse is the question. The S5 theory `GoedelVariantHOML2` already proves

$$\text{MC} : \quad [\varphi \supset \Box \varphi],$$

from Ax2a, Ax2b, Th5, the definition of God-likeness, and symmetry of R [7]. Applied to the world-proposition “the world is w ”, that lemma already confines R to the identity. The same theory checks consistency only by `nitpick[satisfy, card=1]`.

The question comes from a return to the footnote. Christoph Benzmüller and Dana Scott, in the *Monatshefte für Mathematik* [8], and in the matching Archive of Formal Proofs entry [7],

study Gödel’s own generalized conjunction as an axiom they call Ax1Gen. Positivity of being God-like is not assumed. It is lemma L, derived from Ax1Gen. Essence keeps Scott’s extra conjunct. Individual quantifiers in the argument are actualist: an existence predicate says who is around at a world. In their Figure 7, Theorem Th3 is

$$[\Diamond(\exists^E x. Gx) \supset \Box(\exists^E y. Gy)].$$

In the S5 embedding the proof uses symmetry of the accessibility relation and nothing else from S5. The theory `GoedelVariantHOML2inS4` repeats the same non-logical axioms over an S4 frame (reflexive and transitive, symmetry not assumed) and leaves Th3 as `oops`, with the comment “Open problem”. The header of that theory says that the proof of Th3 fails in S4. Figure 8, which repairs the argument by changing necessary inclusion instead of essence, has the same open `oops` in `GoedelVariantHOML3inS4`. The authors’ formalization reports neither an S4 proof nor an S4 countermodel.

For a Lean 4 transcription of that AFP text, the literal reading proves Th3, and the mechanism is collapse: the axioms force R to be the identity, so the only S4 models are discrete and Th3 holds there trivially. Two reconstructions of Ax1Gen refute Th3 on a two-world S4 chain. Among the three readings examined, with full comprehension, the answer turns on whether both halves of Ax1Gen read Φ at the same world. The domains were not varied, and Figure 8 was not the variable in that comparison. An Isabelle check that would instantiate the collection of Section 4 inside `GoedelVariantHOML2inS4` has not been run: Isabelle was not available.

2 Figure 7, as formalized

Worlds form a type W , individuals a type Ind , and accessibility is a relation R on W . A world-proposition is a map $W \rightarrow \mathbf{Prop}$. A property is a map $Ind \rightarrow W \rightarrow \mathbf{Prop}$, individual first and world second, matching the AFP type $e \Rightarrow i \Rightarrow \mathbf{bool}$. Positivity P is a world-relative predicate on properties. Necessity and possibility are the usual quantifications along R :

$$(\Box\varphi)w \equiv \forall v. R w v \rightarrow \varphi v, \quad (\Diamond\varphi)w \equiv \exists v. R w v \wedge \varphi v.$$

Validity $\llbracket\varphi\rrbracket$ means φ holds at every world.

An existence predicate $ex : Ind \rightarrow W \rightarrow \mathbf{Prop}$ plays the role of the AFP constant *existsAt*. The outer type of individuals is fixed. Who exists may vary. Actualist quantifiers are the restrictions

$$(\forall^E y. \varphi y)w \equiv \forall y. ex y w \rightarrow \varphi y w, \quad (\exists^E y. \varphi y)w \equiv \exists y. ex y w \wedge \varphi y w.$$

Quantifiers over properties stay possibilist: they range over every map $Ind \rightarrow W \rightarrow \mathbf{Prop}$.

God-likeness, necessary inclusion, essence, and necessary existence are the Figure 7 clauses. Writing $\varphi \cdot \psi$ for the pointwise conjunction of properties and $\sim \varphi$ for pointwise negation:

$$\begin{aligned} Gx &\equiv \forall\varphi. P\varphi \supset \varphi x, \\ \varphi \supset_N \psi &\equiv \Box(\forall^E y. \varphi y \supset \psi y), \\ \varphi \text{ Ess. } x &\equiv \varphi x \wedge (\forall\psi. \psi x \supset (\varphi \supset_N \psi)), \\ Ex &\equiv \forall\varphi. (\varphi \text{ Ess. } x) \supset \Box(\exists^E z. \varphi z). \end{aligned}$$

The AFP binder in the last clause shadows x ; the Lean definition uses a fresh variable, which is the same formula. The axioms, all under validity, are Ax1 (binary conjunction), Ax2a (exclusive disjunction of $P\varphi$ and $P(\sim\varphi)$), Ax2b ($P\varphi \supset \Box P\varphi$), Ax3 (PE), and Ax4 (a positive property passes positivity along $\supset_N$). Ax1 is part of the package and holds in the one-world model of Section 4. Th3 does not use it. The AFP proof of Th3 does not use it either.

The footnote becomes two abbreviations and one axiom.

$$\begin{aligned}\text{PosProps } \Phi &\equiv \forall \varphi. \Phi \varphi \supset P \varphi, \\ \text{ConjOfPropsFrom } \varphi \Phi &\equiv \Box(\forall^E z. \varphi z \leftrightarrow (\forall \psi. \Phi \psi \supset \psi z)), \\ \text{Ax1Gen} &\equiv \lfloor (\text{PosProps } \Phi \wedge \text{ConjOfPropsFrom } \varphi \Phi) \supset P \varphi \rfloor.\end{aligned}$$

Both Φ and φ are universally quantified. This is the text of `GoedelVariantHOML2` and of `GoedelVariantHOML2inS4`. Lemma L is $\lfloor P G \rfloor$, obtained by feeding `Ax1Gen` the collection of properties that are positive at the world of evaluation. It is a lemma, not an axiom. Scott’s later A3 postulates that same sentence outright [14]. The two packages are different, and Section 6 is there to keep them apart.

In the S5 theory, `Th3` is proved from `Th2` by the three steps the AFP script writes down: existence of a God-like being implies necessary existence; the diamond of the antecedent is therefore a diamond of a box; symmetry turns that diamond-box back into a box. The Lean theorem `th3_of_symmetric` is that argument, with the witness individual written out. Reflexivity is not used. Lemma `MC` in the same theory is the collapse schema quoted in Section 1. Its script cites `Ax2a`, `Ax2b`, `Th5`, the definition of G , and R_{symm} .

3 What the encoding is

The development is a shallow embedding in Lean 4 [9], with no Mathlib library. Hypotheses are propositions, passed as arguments. There is no Lean `axiom` declaration anywhere in the repository. The host logic is the ordinary Lean kernel. Where a proof uses classical reasoning, `#print axioms` says so (Section 7). Isabelle/HOL is classical [13], so that use of classical logic matches the host logic of the AFP entry. It is still a different host.

The departures from the AFP text are these.

AFP	This Lean development
Isabelle/HOL <code>bool</code>	Lean <code>Prop</code> . Classical steps are exactly the ones whose axiom list contains <code>Classical.choice</code> .
$\lfloor \psi \rfloor \equiv \forall w. \psi w$	The definition <code>valid</code> .
$\text{existsAt} : e \Rightarrow \sigma$	$\text{ex} : \text{Ind} \rightarrow W \rightarrow \text{Prop}$. A fixed outer type, with existence varying by world, is how the AFP file represents varying domains.
Property quantifiers are HOL $\forall$ over functions $e \Rightarrow \sigma$	Lean $\forall$ over functions $\text{Ind} \rightarrow W \rightarrow \text{Prop}$. Comprehension is full, as in their impredicative <code>Ax1Gen</code> .
Figure 8 writes $\varphi \neq (\lambda x. \perp)$, which is extensional inequality of HOL functions	<code>neBot</code> : $\neg \forall x w. \varphi x w \leftrightarrow \text{False}$. The symmetry proof supplies a pointwise witness. Lean’s intensional $\varphi \neq \perp$ is a different statement, and it is not the one used.
Deep surface syntax, Leibniz equality inside the object logic, Nitpick	Absent. The one-world model of Section 4 is the cardinality-one shape Nitpick reported, rebuilt by hand.

World-relative P , actualist individual quantifiers, possibilist quantifiers over properties, Figure 7 essence with the conjunct φx , and the derivation of $P(G)$ are the AFP reading. They are not extra deviations. Older modules in the same repository use a different argument order for properties and a constant domain. Those modules are not the encoding of this paper.

One consequence of Lean’s `Prop` is worth saying before the proofs. A proof of a negation may be a classical contradiction rather than a construction of a counter-witness. The symmetry argument is of that kind. The countermodel of Section 5 is not: failure of `Th3` at the source is an explicit world, an explicit individual, and a computation.

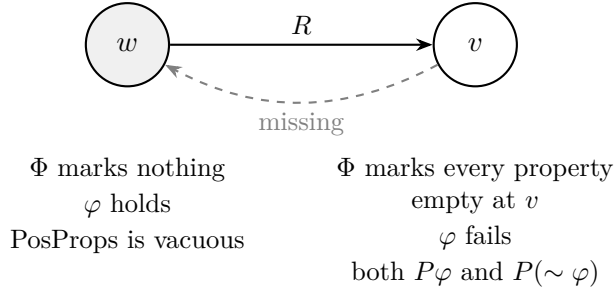


Figure 1: The collection used by `fig7_implies_symmetric`. At the source, Φ is empty, so Ax1Gen owes nobody a positivity premise. At a successor that cannot see the source, the same Φ marks every property empty at u , φ is “the world is the source”, and Ax4 with Ax2a disagree.

The check that could still fail is not another Lean proof. It is the Isabelle instantiation

$$\text{Ax1Gen}[\text{of } “\lambda\psi u. u \neq w \wedge (\forall x. \neg\psi x u)” “\lambda x u. u = w”]$$

inside `GoedelVariantHOML2inS4`. Isabelle was not available, so that check has not been run.

4 The literal axiom forces the identity

4.1 The collection

Ax1Gen invites a definition of a property by the collection it conjoins. The abbreviation does not require the collection to be the same collection at every world. PosProps Φ is evaluated at the world where the axiom is applied. ConjOfPropsFrom puts Φ under a box, so on the right-hand side Φ is read at successor worlds.

Fix a world w and define

$$\begin{aligned}\varphi &::= \lambda_u. (u = w), \\ \Phi &::= \lambda\psi u. (u \neq w \wedge \forall x. \neg\psi x u).\end{aligned}$$

At w , the clause $u \neq w$ fails, so Φ marks no property. Positivity of “every marked property” is vacuous, and Ax1Gen still concludes that φ is positive at w . At a successor $u \neq w$, Φ marks every property empty at u . The right-hand side of the biconditional asks such a property of an individual and cannot hold, and the left-hand side, “the world is w ”, is false. Both sides fail together. Positivity, once granted at w , is pushed along an edge by Ax2b. At a far world v that cannot see w , necessary inclusion of φ in $\sim\varphi$ is vacuous, because the antecedent never fires. Ax4 passes positivity to the negation. Ax2a refuses to let a property and its negation both be positive.

Lemma 1 (`pos_of_home`). *From Ax1Gen alone: if φ holds of every individual that exists at w whenever Rww , then $P\varphi$ holds at w .*

Sketch. Instantiate Ax1Gen at the collection $\Phi\psi u \equiv (u \neq w \wedge \psi = \varphi)$. At w that collection marks nothing, so PosProps is vacuous. At a successor equal to w , the hypothesis supplies φ of every existent, which is what the biconditional asks. At a successor other than w , the only marked property is φ itself, so the biconditional is the identity. Ax1Gen returns $P\varphi$ at w . $\square$

The collection of Figure 1 is the case in which φ says “the world is w ”. Lemma 1 is the same trick for an arbitrary property. It is what ties positivity to who happens to exist at w .

4.2 Reflexivity, the B schema, and the identity

Theorem 2 (`refl_of_gen`). *From $Ax1Gen$ and $Ax2a$, R is reflexive.*

Sketch. If $\neg R w w$, the hypothesis of Lemma 1 is vacuous for every property. Both $\perp$ and $\sim \perp$ are therefore positive at w . $Ax2a$ forbids that pair. $\square$

Theorem 3 (`B_schema`). *From $Ax1Gen$, $Ax2a$, $Ax2b$, and $Ax4$, every world-proposition q satisfies $q \supset \Box \Diamond q$.*

Sketch. Suppose q holds at w , and $R w v$. Let φ ignore the individual and hold at a world iff q does. Lemma 1 makes φ positive at w : under reflexivity, every existent at w has φ , because q is true there. $Ax2b$ pushes positivity to v . If v saw no world at which q holds, then φ would necessarily include $\perp$ at v . $Ax4$ would make $\perp$ positive, and a second use of $Ax4$ would make $\sim \perp$ positive. $Ax2a$ forbids that pair. So v sees some world at which q holds. $\square$

Comprehension is used: the property φ is the map $\lambda_u. q u$, and “the world is w ” is a world-proposition of the same kind. Theorem 4 is the case $q \equiv (u = w)$.

Theorem 4 (`fig7_implies_symmetric`). *From $Ax1Gen$, $Ax2a$, $Ax2b$, and $Ax4$ it follows that R is symmetric. Reflexivity and transitivity are not hypotheses of this implication. Reflexivity is already a consequence of $Ax1Gen$ and $Ax2a$ (Theorem 2).*

Sketch. Fix $R w v$, and suppose $\neg R v w$. Use the collection of Section 4.1. $Ax1Gen$ yields $P \varphi$ at w , and $Ax2b$ pushes it to v . At v , necessary inclusion of φ in $\sim \varphi$ is vacuous, because a successor of v cannot be w . $Ax4$ yields $P(\sim \varphi)$ at v , contradicting $Ax2a$. Hence $R v w$. $\square$

$Ax3$ is not used for symmetry. The same four hypotheses are unsatisfiable on the two-world chain of Figure 2. That fact is `fig7_unsat_on_S4_chain`. The audit module restates it as `literal_unsat_on_R_chain`. That chain is reflexive, transitive, and not symmetric. It is not a countermodel to literal Figure 7, because no positivity and no existence predicate satisfy the axioms there.

Theorem 5 (`R_is_identity`). *From $Ax1Gen$, $Ax2a$, $Ax2b$, $Ax3$, and $Ax4$, $R w v$ if and only if $v = w$. Symmetry is not a hypothesis.*

Sketch. Theorem 2 gives the right-to-left direction. For the other direction, let φ be “the world is w ”. Lemma 1 makes it positive at w , and $Ax2b$ carries that positivity to any v with $R w v$. Theorem Th5, which is necessary existence of a God-like being and which the AFP derives from Th3 and Th4, supplies at v an existing individual who is God-like. God-likeness means every positive property holds of that individual, so the world v is w . The Lean proof of Th5 goes through `th3_fig7`, which uses symmetry. The symmetry it uses is Theorem 4, derived from the same axioms, not an extra frame assumption. $\square$

Theorem 6 (`MC`). *From the same five axioms, every world-proposition q satisfies $[q \supset \Box q]$.*

Sketch. If q holds at w and $R w v$, Theorem 5 forces $v = w$, so q holds at v . $\square$

This is the schema of the AFP lemma `MC`. The AFP proof assumes R_{symm} . Theorem 6 does not. The discrete-frame consequence is the same one their lemma already had, once symmetry was known: applied to “the world is w ”, collapse leaves only the reflexive loop. What the literal S4 problem was missing is that symmetry does not have to be assumed. The axioms produce it, and they produce the identity relation.

Theorem 7 (`th3_fig7`). *From $Ax1Gen$, $Ax2a$, $Ax2b$, $Ax3$, and $Ax4$,*

$$[\Diamond(\exists^E x. Gx) \supset \Box(\exists^E y. Gy)].$$

The literal axioms force R to be the identity. Their S4 models are discrete, and Th3 holds there trivially: a diamond and a box along the identity relation are the same quantifier. The earlier route to the same theorem derived symmetry and then repeated the AFP three-step argument (`th3_of_symmetric`). That argument is still in the file. It is not an S4 theorem with room in it. There is no non-discrete S4 model in which the antecedent can hold while the consequent fails, because there is no non-discrete model of the axioms at all.

The intermediate AFP lemmas that do not mention symmetry remain rediscoveries: `lemma_L`, `ax2b'`, `th1_fig7`, `th2_fig7`, `th4_fig7`, and `th5_fig7`. The one-world control shows that the axioms are not jointly absurd on a discrete frame.

Proposition 8 (`unit_fig7_th3`). *On a single world, with the identity relation, one individual existing there, and principal positivity $P\varphi \equiv \varphi()()$, axioms $Ax1$, $Ax2a$, $Ax2b$, $Ax3$, $Ax4$, and $Ax1Gen$ hold, and Th3 holds.*

This is the shape Nitpick reported at cardinality one, checked directly. It is a consistency witness for the literal package. It is not a claim that the axioms are true of anything outside the model. On that same one-world identity frame the literal axiom, the boxed reading, and the world-invariant reading coincide (`readings_coincide_on_unit`; no kernel axioms). The two spellings of the boxed reading coincide on every frame (`ax1GenBox_iff_inBox`).

4.3 Figure 8

Figure 8 (`GoedelVariantHOML3`) changes two clauses. Necessary inclusion gains a side condition, and essence loses the conjunct φx :

$$\varphi \supset_N \psi \equiv \Box(\varphi \neq (\lambda x. \perp) \wedge \forall^E y. \varphi y \supset \psi y),$$

with essence the universal closure of that inclusion under properties the individual has. In the Lean file the inequality is `neBot`, the extensional disagreement of Section 3. The S4 theory again leaves Th3 as `oops`, comment “Open problem”. Figure 8 was not varied as part of the comparison in Section 5. The two theorems here are further facts about the literal axiom.

Theorem 9 (`th3_fig8`). *From literal $Ax1Gen$, $Ax2a$, $Ax2b$, Figure 8’s $Ax3$, and Figure 8’s $Ax4$, Th3 holds.*

The symmetry argument is Theorem 4. Figure 8’s inclusion asks, in addition, that φ not be identically false. The witness property “the world is w ” disagrees with $\perp$ at w , once any individual is on hand to point at. Theorem `th3_fig8` takes that individual from the diamond in the antecedent of Th3, proves symmetry from it, and only then runs the back-edge. On the S4 chain, once an individual is given, the Figure 8 axioms used for symmetry are unsatisfiable (`fig8_unsat_on_S4_chain`). The same emptiness remark applies: Theorem 5 did not have to be re-proved for Figure 8’s inclusion, and this note does not claim that the Figure 8 package forces the identity relation.

Theorem 10 (`th4_fig8`). *From literal $Ax1Gen$ and $Ax2a$, $\lfloor \Diamond(\exists^E x. Gx) \rfloor$.*

Sketch. Lemma L makes G positive at w . Theorem 2 makes w see itself. If w saw no existent God-like being, then $\sim G$ would hold of every existent at w . Lemma 1 would make $\sim G$ positive, contradicting $Ax2a$. $\square$

The proof does not use $Ax4$ and does not use Figure 8’s side condition on inclusion. In `GoedelVariantHOML3.thy` the Th4 script is `oops`, with a comment that `sledgehammer` found a proof from $Ax2a$, lemma L, and $Ax1Gen$, and Th4 is then reintroduced by `axiomatization`. Theorem 10 is that missing proof. A Figure 8 Th5 is not claimed.

4.4 The literal axiom is not the footnote

Theorem 7 is a theorem of the literal AFP axiom. By Gödel’s own gloss, that axiom is not his.

The footnote says that a conjunction of positive properties is positive [11, 4]. The gloss printed with the proof says that positive means positive in the moral aesthetic sense, independently of the accidental structure of the world [11]. The AFP abbreviation does something else. Its type lets Φ vary by world. Lemma 1 then says that Ax1Gen alone makes every property true of all existents at w positive at w , whenever w sees itself, and makes every property positive at w if w does not see itself. Positivity at w tracks who exists at w . That is the accidental structure the gloss excludes.

The split is what a shallow embedding does if PosProps is left outside a box while Φ still depends on the world. It is the text of `GoedelVariantHOML2`, faithfully transcribed. It is not a misprint, and it is not a reading of the footnote. The footnote does not mention worlds. Theorem 7 is a genuine theorem about an axiom that, by that gloss, is not Gödel’s.

5 Two reconstructions

Everything in Figure 7 stays put except Ax1Gen. There are two replacements. Neither is dictated by the footnote. Both are reconstructions. What both remove is Φ being read at two different worlds by the two halves of the axiom. An earlier draft called the first the closer repair. That ranking is dropped. One may prefer the second because “any number of summands” is a single collection, and a single collection does not change with the world. The footnote still does not say so. Rigidity, in the sense of Ax2b, is kept in all three readings. The Lean name `Ax1GenRigid` means that Φ is world-invariant. It does not mean that positivity is rigid.

R1, positivity in the box. Both halves are judged at the same accessible worlds:

$$[\Box(\text{PosProps } \Phi \wedge \forall^{Ez}. \varphi z \leftrightarrow (\forall \psi. \Phi \psi \supset \psi z)) \supset P \varphi].$$

Call this `Ax1GenInBox`. In K, a box of a conjunction is a conjunction of boxes, and `ConjOfPropsFrom` is already a box, so the displayed form is equivalent to

$$[(\Box(\text{PosProps } \Phi) \wedge \text{ConjOfPropsFrom } \varphi \Phi) \supset P \varphi],$$

which is `Ax1GenBox`. The equivalence `ax1GenBox_iff_inBox` depends on no kernel axioms. One countermodel covers both spellings.

R2, a world-invariant collection. Restrict Ax1Gen to collections that do not depend on the world: $\Phi \psi w \leftrightarrow \Phi \psi v$ for every property and every pair of worlds. PosProps and `ConjOfPropsFrom` stay literal. Call this `Ax1GenRigid`. R2 moves no box. It freezes Φ . The collection of Figure 1 is not world-invariant, so R2 does not apply to it.

5.1 Lemma L still derives

Under R1, lemma L is immediate from the repaired axiom alone (`lemma_L_box`, `lemma_L_inBox`). Take Φ to be P itself. At every world, every property that Φ marks is positive there, by identity, so the extra box is free. The conjunction of whatever is positive at a world is G at that world, by the definition of G .

Under R2 the same Φ may be refused, because P itself need not be world-invariant. Take a snapshot at the world w where the lemma is being proved: $\Phi \psi _ \equiv P \psi w$. That collection ignores its world argument. At a successor v , the conjunction it defines is G only if positivity at w and positivity at v agree. Ax2a and Ax2b give that agreement along an edge (`pos_agree`): a positive property stays positive by Ax2b, and a property positive at v cannot have a positive

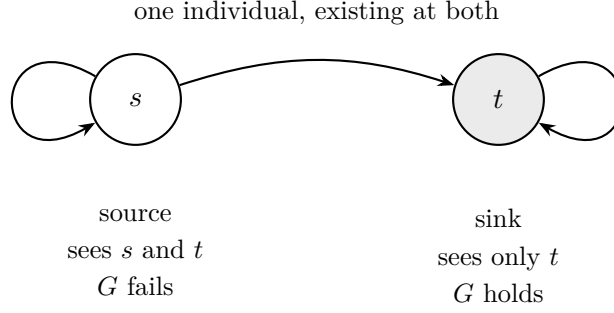


Figure 2: The S4 chain of `R_chain`, drawn as source $s = \text{false}$ and sink $t = \text{true}$. Reflexive, transitive, and not symmetric. The domain has one element, and existence is true at both worlds. Under `chainP`, a God-like being exists at the sink only, so at the source a God-like being is possible and not necessary. Literal `Ax1Gen` has no model on this frame. Both reconstructions do.

negation at w , or `Ax2b` would carry the negation forward and break `Ax2a` at v . So `lemma_L_rigid` needs `Ax2a` and `Ax2b`. Literal `lemma_L` needed neither. The box in `R1` re-reads $\Phi := P$ at the successor rather than freezing it at the source.

Symmetry is not a consequence of either reconstruction. The proof is the chain in Figure 2 and Theorem 12: the frame is not symmetric, and each reconstruction holds on it. The collection in Theorem 4 does not witness a contradiction under either reconstruction. It is empty at the source, so source `PosProps` holds, but at a successor it marks every property empty at that world, and in particular it marks $\perp$. `R1`'s boxed `PosProps` demands $P(\perp)$ there. `R2` does not apply, because the collection is not world-invariant. That is why the derivation of a back-edge has nothing left to instantiate. It is not itself the proof that symmetry can fail.

5.2 The chain

Let the worlds be the two Booleans. The source `false` sees both worlds. The sink `true` sees only itself (Figure 2). The relation is reflexive and transitive. It is not symmetric: the sink does not see the source (`R_chain_not_symmetric`). The domain has one element, and that individual exists at both worlds. Existence is the constant `true` predicate. The actualist restriction never excludes anyone, so this countermodel does not exercise the actualist machinery. Positivity ignores the world at which it is asked and reads the sink:

$$P\varphi \equiv \varphi() \text{ true.}$$

A property is positive precisely when the individual has it at t .

At the sink, God-likeness is a tautology. G says that every positive property holds of the individual there, and positivity means that the property holds of the individual there (`chain_god_sink`; no kernel axioms). At the source it fails. The property “the world is the sink” holds at the sink, hence is positive, and the individual does not have it at the source (`chain_not_god_source`; no kernel axioms).

Theorem 11 (`chain_th3_fails_at_source`). *At the source, $\Diamond(\exists^E G)$ holds and $\Box(\exists^E G)$ fails.*

Sketch. The source sees the sink, where the individual exists and is God-like. The source also sees itself, where that same individual exists and is not God-like. The diamond is the first fact. The failure of the box is the second. $\square$

No kernel axiom is used. The packaged denial `chain_not_Th3` is the same fact pulled back along the definition of `Th3`. `chain_not_necessary_existence` is the failure of the box as a validity, equally axiom-free.

Reading of Ax1Gen	Lemma $P(G)$	What R must be	Th3 in S4
Literal AFP	from Ax1Gen alone	the identity (Thm. 5)	holds, on discrete frames
R1, boxed	from R1 alone	not symmetric (Thm. 12)	fails (Thm. 12)
R2, world-invariant Φ	needs Ax2a and Ax2b	not symmetric (Thm. 12)	fails (Thm. 12)

Table 1: Three readings of where Ax1Gen reads Φ . The literal row uses Ax3 for the identity. The witness for both negative cells is Figure 2. On a one-world frame the three readings coincide (Proposition 13).

Theorem 12 (`r1_s4_countermodel`, `r2_s4_countermodel`). *On this frame the relation is an S4 frame and not a symmetric one. Ax1, Ax2a, Ax2b, Ax3, Ax4, and both spellings of R1 hold, and Th3 fails. The same frame, the same positivity, and the same existence predicate satisfy R2, and Th3 fails.*

The axiom checks are direct, and most of them use no kernel axioms. Ax2a is the exclusive-or of “ φ holds at the sink” and “ φ fails at the sink”, which is classical excluded middle (`chain_ax2a`). Ax2b is immediate because P does not look at its world argument: positivity is already global. Ax4 follows because both worlds see the sink, so a necessary inclusion can be read there, where P evaluates extensions. Ax3 holds because P on necessary existence only asks for the extension at the sink, and the essence hypothesis at the sink includes that the property holds of the individual there. For R1, the boxed biconditional at whichever world one starts from can be read at the sink, which both worlds see, and the right-hand side at the sink is exactly the extension P consults. For R2 the same computation works, and world-invariance lets a mark at the sink be copied back to the world where PosProps was assumed.

Proposition 13 (`literal_to_R1`, `literal_to_R2`, `readings_coincide_on_unit`). *Literal Ax1Gen implies R1 on every reflexive frame, and implies R2 on every frame. On a one-world identity frame the three readings coincide. On the chain, this positivity satisfies both reconstructions and does not satisfy literal Ax1Gen.*

One direction is `literal_to_R1`. The other is `literal_to_R2`. The one-world coincidence is `readings_coincide_on_unit`. Every collection on a one-world frame is world-invariant, and along the identity relation a box does not leave the world. If this positivity satisfied literal Ax1Gen, Theorem 4 would make the chain symmetric. `chain_not_literal_Ax1Gen` records that it does not. The converse implications, reconstruction implies literal, are not proved in general. They are proved on the one-world frame, and the chain separates the readings in the other direction: the reconstructions have a model there and the literal axiom has none.

Table 1 is the comparison among these three readings. The possibilist twin also leaves Th3 as `oops`. It is not encoded. Its AFP name is `GoedelVariantHOML2possInS4`. Domains were not varied: the outer type of individuals stays fixed, and the chain uses a one-element domain with existence always true. A constant domain with no existence predicate would be a different theorem, and the Scott-style chain already studied in this repository is that different theorem (Section 6).

6 What was already known

Several nearby statements are in the same repository, and they were already theorems of other people. They are named here so that the results above are not asked to carry them.

Sobel showed that Scott-style premises yield $\varphi \supset \Box\varphi$ [15, 16]. Benz Müller and Fuenmayor encoded Scott’s, Anderson’s, and Fitting’s variants and checked that Scott’s intensional positivity collapses modality, while Anderson’s emendation and Fitting’s extensional positivity do not [5, 1, 2, 10]. The repository’s collapse modules rediscover the Scott-side collapse, including the

world-relative variant in which collapse along R is separated from Sobel’s global biconditional when R is not universal. That rediscovery is not Theorem 6. Theorem 6 is collapse for the literal Figure 7 package, and the prior art for that schema is the AFP lemma MC, which assumes symmetry [7].

Kanckos and Woltzenlogel Paleo showed that Scott’s argument goes through in KB: symmetry suffices, and S5 is more than the proof uses [12]. Benzmlüller and Scott make the same point for their Th3, which cites R_{symm} and not reflexivity or transitivity [8, 7]. The repository’s Scott-style development rediscovers the positive half (local Th3 from symmetry alone, on that encoding) and pairs it with countermodels once symmetry is dropped. Those countermodels postulate positivity of being God-like. Figure 7 derives that sentence as lemma L. A countermodel that assumes the lemma does not answer a question whose point is what happens when the lemma is earned from Ax1Gen. The file `S4open.lean` in the repository is an explicit record of this mismatch. Theorem 12 is not that countermodel. On the chain of Figure 2 the literal axiom has no model at all.

Anderson’s emendation—only one direction of the exclusion axiom, and God-likeness as possession of exactly the positive properties, necessarily—is a known repair [1, 2]. A thin fragment of it is formalized elsewhere in the repository, and a two-world witness there keeps a contingent fact alive. That fragment is a rediscovery of Anderson’s repair. Fitting’s extensional block is not formalized [10]. Neither is part of Theorems 5 and 12.

The early mechanizations of the Scott-style argument [6] are the same Scott package. Benzmlüller’s note on ultrafilters [4] derives $\lceil \Box(\exists^E x. Gx) \rceil$ already in K, from an ultrafilter axiom U1, a conjunction principle, an entailment axiom, and the definition of G . The note is explicit that this is a simplified package, and that Scott’s own A3 postulated $P(G)$ in place of the footnote’s conjunction. It is not Figure 7’s axiom list, and it does not use the AFP scoping of Ax1Gen. The Archive entry on simplified variants [3] studies arguments that avoid essence and necessary existence and that are valid in K or KT. Those are different axioms again.

What was checked, for the claim that the identity proof and the two reconstructions were not found already, is the following AFP theories, together with the S4 and S5 embeddings they import:

- `GoedelVariantHOML2` and `GoedelVariantHOML2inS4` (Figure 7, S5 and S4);
- `GoedelVariantHOML3` and `GoedelVariantHOML3inS4` (Figure 8, S5 and S4);
- `GoedelVariantHOML2possInS4` (the possibilist S4 twin, not encoded here).

The literature named in this section was checked as well, namely the items cited from the repository notes that accompany the formalization: Sobel [15, 16], Anderson [1], Anderson and Gettings [2], Fitting [10], Kanckos and Woltzenlogel Paleo [12], Benzmlüller and Woltzenlogel Paleo [6], Benzmlüller and Fuenmayor [5], Benzmlüller [4], and the simplified-argument AFP entry [3]. In `GoedelVariantHOML2`, lemma MC is prior art for collapse under symmetry, and Th3 is proved from symmetry. In the AFP S4 files, Th3 is the open `oops`. In [4] the K-theorem is the ultrafilter package just described. No proof of the identity, or of collapse, for literal Figure 7 without a symmetry hypothesis, and no countermodel of the two reconstructions, turned up in those sources. That is a statement about where the search went. It is not a priority claim.

7 What the kernel accepts

The development builds with `lake build` on Lean 4.34.0, toolchain `leanprover/lean4:v4.34.0`, in 37 jobs, with no Mathlib. A search of the Lean sources finds no `sorry`, no `native_decide`, and no `axiom` command. The strings that look like those words sit in comments. Hypotheses of the ontological argument are parameters of type `Prop`.

Table 2 is the output of `#print axioms` on the new lemmas and on the headline theorems, run in that toolchain after the build. “None” means Lean reported that the constant

Theorem	#print axioms
pos_of_home, refl_of_gen, B_schema, R_is_identity, MC, th4_fig8	propext, Classical.choice, Quot.sound
literal_to_R1, literal_to_R2, R1_to_literal_of_identity, literal_iff_R1_of_identity, R2_to_literal_of_one_world, readings_coincide_on_unit	none
literal_unsat_on_R_chain	propext, Classical.choice, Quot.sound
chain_not_necessary_existence	none
lemma_L, ax2b', th1_fig7, th2_fig7, th3_of_symmetric	none
th1_fig8, th2_fig8, th3_fig8_of_symmetric	none
fig7_implies_symmetric, th3_fig7, fig7_unsat_on_S4_chain, th4_fig7, th5_fig7	propext, Classical.choice, Quot.sound
th3_fig8, fig8_implies_symmetric, fig8_unsat_on_S4_chain	propext, Classical.choice, Quot.sound
unit_fig7_th3	propext, Classical.choice, Quot.sound
ax1GenBox_iff_inBox, lemma_L_box, lemma_L_inBox, lemma_L_rigid, pos_agree	none
chain_th3_fails_at_source, chain_not_Th3, chain_god_sink, chain_not_god_source, chain_ax1GenRigid	none
R_chain_transitive, R_chain_not_symmetric	propext
chain_not_literal_Ax1Gen	propext, Classical.choice, Quot.sound
r1_s4_countermodel, r2_s4_countermodel	propext, Classical.choice, Quot.sound

Table 2: Kernel axioms of the audit lemmas and the headline theorems, Lean 4.34.0. The repaired countermodels inherit **propext** from the chain’s frame facts and **Classical.choice** from Ax2a. The failure of Th3 at the source, taken by itself, depends on none of them.

does not depend on any axioms. The triple **propext**, **Classical.choice**, **Quot.sound** is the standard classical package of the Lean kernel: propositional extensionality, choice, and the soundness of quotients that the classical development uses. **Classical.choice** enters through **Classical.byContradiction** in the symmetry argument, in reflexivity, in the B schema, in the identity, in collapse, and in both proofs of possible existence, and through **Classical.em** in the one-world check of Ax2a and in **chain_ax2a**. **propext** on its own is the chain’s transitivity and the proof that the chain is not symmetric, both of which simplify propositional equalities. No custom axiom appears.

8 Conclusion

Theorem 5 says that the literal Figure 7 axioms force R to be the identity, and Theorem 6 says that modal collapse follows, with no symmetry hypothesis. The AFP lemma MC is the same schema under R_{symm} . The mechanism is Lemma 1: Ax1Gen, read so that the two halves consult Φ at different worlds, makes positivity follow the existents at the source. The literal axioms force R to be the identity; their S4 models are discrete, and Th3 holds there trivially. Theorem 12 says that two reconstructions, on which both halves read Φ at the same world, still yield lemma L and fail on a two-world S4 chain. Among the three readings examined, with full comprehension, the answer turns on whether both halves of Ax1Gen read Φ at the same world. Domains and Figure 8 were not varied. The chain that separates the reconstructions has a one-element domain and existence always true.

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kernel-checked in the repository <https://github.com/drewarrowood/godel-ontological>.

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